GENERATION SHIP

A cross-course activity

Game Master Handbook · Course offering for the German Junior Academies

A course offering within the German Junior Academies (Deutsche Junior Akademien), prepared by a participant

In cooperation with the Westphalian Public Observatory and Planetarium Recklinghausen (Westfälische Volkssternwarte und Planetarium Recklinghausen)

[Insert the observatory's logo here — please request the official logo at sternwarte-recklinghausen.de or directly from the observatory]

1. Overview

“Generation Ship” is a cross-course activity prepared as an independent course offering within the German Junior Academies (Deutsche Junior Akademien) — in cooperation with the Westphalian Public Observatory and Planetarium Recklinghausen. Participants take on different roles aboard a spaceship setting out for Proxima Centauri, with an assumed travel time of 60,000 years. Since this timespan far exceeds a single human lifetime, the mission has to be planned as a multi-generational project: together, participants develop a concept for how knowledge is preserved, how conflict is avoided, and how the mission can succeed despite conflicting interests — and they draw up a blueprint for the ship.

The activity combines astronomy/physics (travel times, distances, propulsion concepts) with social and political science, ethics, biology and engineering, making it especially well suited as a cross-course offering.

2. Learning Objectives

- Interdisciplinary thinking: connecting astronomical facts with social, ethical and technical questions
- Perspective-taking: arguing from an assigned role, even where it differs from one's own opinion
- Negotiating and finding compromises under genuine conflicts of interest
- Systems thinking: assessing the long-term consequences of decisions across many generations
- Creativity and spatial thinking in planning the ship

3. Key Facts

At a glance

Target audience: participants of the German Junior Academies, cross-course

Roles per group: 3 mandatory roles (each in Variant A or B) + up to 5 of 10 optional roles (see role cards)

Recommended group size: 4–8 people; with more participants, run several groups in parallel

Time required: approx. 3–4 hours, can be split across several sessions

Materials: role cards, blueprint (3 pages per group), player booklet, pens, optionally a poster/flipchart, a timer

4. Starting Situation (read-aloud text for the introduction)

You are on the verge of humanity's greatest departure yet. After decades of preparation, the moment has finally come: a generation ship is about to set out for Proxima Centauri, the star nearest to our Sun, about 4.2 light-years away. With propulsion technology conceivable today, the journey would take roughly 60,000 years — more than 2,000 generations. None of you will live to see the arrival. And yet today you must decide how this journey can succeed: How does knowledge survive for millennia? How do you prevent the onboard society from falling apart, or from forgetting why it set out in the first place? And what does the ship itself need to look like for people to live in it across millennia? Each of you represents your own interests here — and they are not always compatible with one another.

Note for the game master: the figure of 60,000 years was chosen deliberately (the observatory recommended 30,000–50,000 years for comparable formats; here it was rounded up to 60,000 to make the multi-generational character even clearer). If you like, you can work out roughly how this number comes about together with the group beforehand (distance to Proxima Centauri ≈ 4.2 light-years; speed of present-day probes such as Voyager 1 ≈ 17 km/s → travel time well over 70,000 years; faster but still untested concepts would bring it down to a few millennia). This makes for a short astronomy-based introduction, e.g. with the observatory's support.

5. Assigning Roles

Every group has three mandatory roles that carry the game's core conflict and therefore always need to be filled. For each mandatory role there is also an equivalent alternate role with the same function within the conflict, but a different angle on the topic — so you don't have to hand out the same three roles to every group:

- Society & justice: Social Scientist — or alternatively — Civil Rights Advocate / Elected Crew Representative
- Power, control & resources: Investor — or alternatively — Chief of Security / Military Commander
- Technology & knowledge: Chief Engineer — or alternatively — Head of AI & Systems Management

Each group uses exactly one variant from each of the three pairs — this way the core conflict (participation vs. control vs. technical feasibility) is always preserved, even when several groups play at the same time.

Beyond that, there are ten optional roles that bring in further perspectives (among others: genetics/medicine, psychology, history/culture, ethics/philosophy, ecology/life support, education, politics/constitutional law, religion/spirituality, law, IT/data security). Up to five of these are freely assigned per group — which is also ideal for not leaving anyone without a role when the number of participants is odd: anyone who no longer fits into their original group simply joins another group with an optional role. Thanks to the larger pool, you also won't have to hand out the same cards every time you run the activity again.

All role cards include a short background for the character as well as their goal for the mission, and use gender-inclusive language throughout. Participants should argue from their role's perspective even where it differs from their own gut feeling — that is exactly where the learning effect lies.

6. Schedule

Phase 0 — Astronomy Introduction (optional, 15–20 min.)

A short input talk on Proxima Centauri, light-years and propulsion concepts — e.g. given by the observatory itself or based on their material. Works well as a shared kickoff before group work begins.

Phase 1 — Introduction & Role Assignment (15–20 min.)

Read out the starting situation (see Chapter 4), split into groups, hand out role cards.

Phase 2 — Preparing Roles (15–20 min.)

Each person reads their own role card and jots down 2–3 arguments, in note form, that they want to bring into the negotiation.

Phase 3 — Negotiation Round: Core Principles (45–60 min.)

The group agrees on basic principles for life on board: How are decisions made? How is knowledge preserved? How are conflicts resolved? Who retains how much control? Outcome: a short shared concept paper or poster (bullet points are enough).

Phase 4 — Ship Planning: Blueprint (45–60 min.)

Using the player booklet (room overview) and the three-page blueprint, the group plans how the ship is divided up: which rooms go where, and why? The engines are already drawn in; participants fill in the rooms and draw the doors themselves. The results of the previous negotiation round (e.g. lots of communal space vs. lots of storage) should be reflected in the plan.

Phase 5 — Crisis Prompt (optional, 15–20 min.)

The game master introduces an unforeseen event (see Chapter 7) and the group has to respond within a few minutes. Good for getting a stalled discussion moving again.

Phase 6 — Presentation (approx. 8–10 min. per group)

Each group presents its concept and its blueprint and explains how the main role interests were taken into account.

Phase 7 — Debrief & Reflection (20–30 min.)

Step out of role, then a joint evaluation in plenary (see Chapter 8).

7. Scenarios for the Game Master

This chapter gives you prompt cards for two purposes: (a) basic questions to get groups that are stuck talking again, and (b) crisis scenarios for Phase 5 that put an ongoing discussion to the test. The questions are deliberately mixed — from down-to-earth and practical, to morally difficult, to playful and hypothetical. You don't need to ask all of them; pick whatever fits the group and the time you have left.

7.1 Basic Questions (usable at any time)

If a group gets lost in details or stops making progress, simple, concrete follow-up questions often help:

- Where exactly does your drinking water come from, and what happens if the treatment system ever fails?
- Where exactly does your food come from? Is it enough for everyone, even as the population grows?
- What happens on board to elderly people who can no longer work?
- What happens to the chronically ill or people in need of care — who looks after them, and what does that cost in space and resources?
- What happens when someone on board dies? Is there even a designated place or process for that?

7.2 Crisis and Conflict Scenarios

For Phase 5 (or as an interjection into a stuck negotiation round), one or two of the following events can be introduced — best done verbally or on a card:

- A central data archive fails — part of the technical knowledge is at risk of being lost. How does the group react?
- The generator, or the central power supply, fails for several days. Which systems keep running first, and which don't?
- A new generation questions the original mission: “Why should we live for a goal we will never reach?”
- Food resources turn out scarcer than planned — cuts have to be made somewhere.
- The investor suddenly demands that scientific data be sent back to Earth as a priority instead of being used for the mission.
- Two onboard factions disagree over the future form of government and threaten to split the community apart — armed conflict on board becomes a real risk.
- Over the generations, deep-rooted discrimination develops on board between different groups (e.g. by origin, role, or birthplace on the ship). How does the onboard society deal with it?

A note on weapons: in earlier runs of this activity (feedback from the observatory), some groups decided to bring weapons on board — partly with the argument that they could be used to assert themselves against other factions or ships. This should not be ruled out from the start; it is a legitimate decision within the game. What's most interesting is the discussion this triggers: who controls the weapons? What does this mean for trust and safety on board? How easily could this escalate?

7.3 Hypothetical Scenario: First Contact

A lighter, playful scenario for a change of pace, or to round off the negotiation round:

- After several thousand years of travel, the ship receives unmistakable signals of extraterrestrial life nearby. What does the group do? Make contact or not? Who decides that? And what if opinions on board differ?

This question works well with a light touch, but it also raises genuinely serious issues along the way: who is allowed to decide on behalf of the entire onboard society? How do you deal with the unknown, and the risk that comes with it?

8. Reflection and Debrief Questions

To close, outside of the roles, in plenary:

1. What was your group's hardest decision? Why?
2. Whose interests came up shortest in your discussion — and why do you think that was?
3. How did your own opinion differ from your role's opinion?
4. Which of your group's decisions would also apply to problems on Earth today — e.g. intergenerational justice in the context of climate change?
5. Looking back, what would you plan differently?

9. Assessment / Observation Criteria (optional)

If the results are to be compared or acknowledged, the following criteria are useful — it is not about “right” or “wrong”, but about traceability and balance:

- Were all roles, and their interests, visibly reflected in the concept?
- Is the solution internally consistent (e.g. does the room layout match the agreed principles)?
- Was the core tension of “preserving knowledge vs. control vs. fairness” handled in a traceable way?
- How creative and well-reasoned was the group's solution to the crisis prompt?

10. Cooperation Note

This course offering was prepared by a participant of the German Junior Academies, in cooperation with the Westphalian Public Observatory and Planetarium Recklinghausen (Westfälische Volkssternwarte und Planetarium Recklinghausen). Please add an appropriate credit line, and if possible the observatory's official logo, to all materials handed out (handbook, role cards, player booklet, blueprint). Observatory contact: info@sternwarte-recklinghausen.de, www.sternwarte-recklinghausen.de.